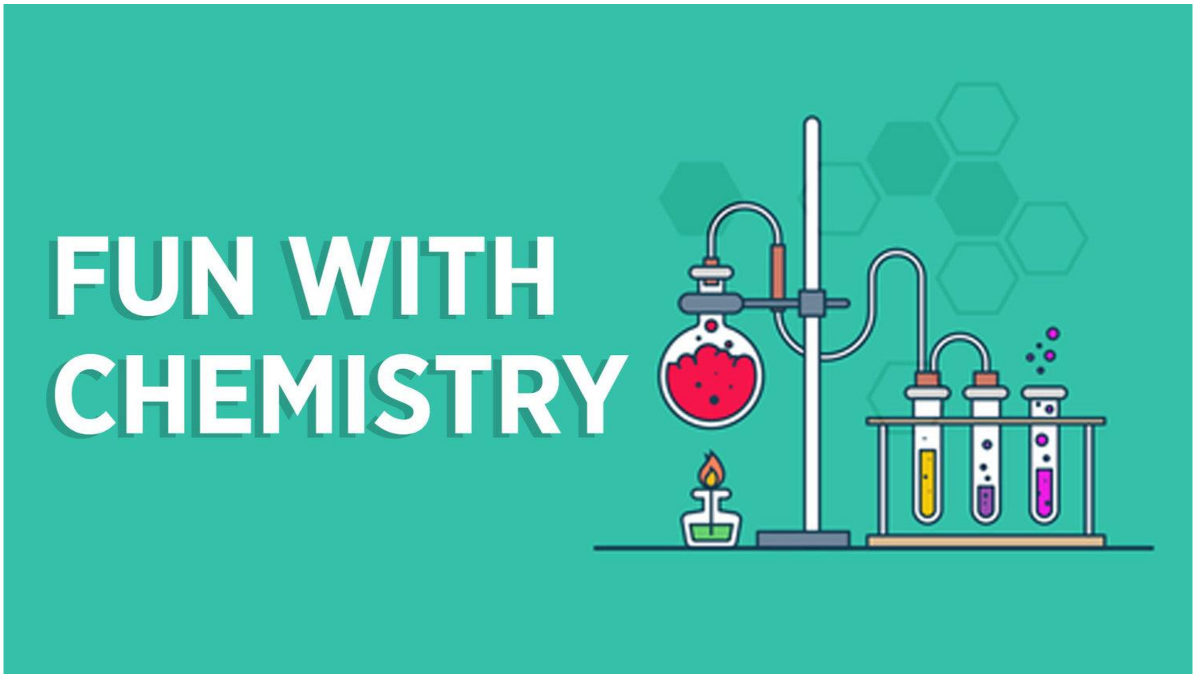
Student Name:

Class period:

AP Chemistry



Unit 1: Chemistry "Fun"damentals

Textbook Reference: Zumdahl Ch1-3

Date	Day	Learning Objectives	Activities	Work on this!
8/13-14	U1D1	Welcome to AP Chem! What do you remember?	Welcome! Day 1 Quiz???	~Syllabus Quiz due next time ~Review- Try pp. 14-18 first. As needed, watch review videos in Schoology **if needed**
8/15-16	U1D2	How many ways to get to a mole?	The Mole in Review PreLab: Baking Soda Mixture	~Mole Reviews in Schoology **if needed!** ~Keep working on Year 1 Review
8/19-20	U1D3	How do I analyze a mixture? What are empirical and molecular formulae?	Mixtures- KI Problem Lab: Baking Soda Mixture Notes: Empirical and Molecular Formula, Percent Composition	~Finish lab if needed ~Review quiz next time! ~Need to rewatch this lesson? Videos in Schoology
8/21-22	U1D4	How can a hydrate formula be experimentally determined?	Year 1 Review Quiz Notes: Hydrates	~Need to rewatch this lesson? Videos in Schoology
8/23-26	U1D5	What is combustion analysis?	Notes: Combustion Analysis Practice Composition Problems	~Need to rewatch this lesson? Videos in Schoology ~Practice Test Questions Posted
8/27-28	U1D6	How are amounts determined in chemistry?	Review Hydrate Lab	~Complete lab if needed ~Study, study, study!
8/29-30	U1D7	What have we learned about Chemistry Fundamentals?	Unit 1 Exam	~Take a rest before Unit 2

AP Chemistry Unit 1 Objectives

1.1	**Moles and Molar Mass**
1.1.A:	Calculate quantities of a substance or its relative number of particles using dimensional analysis and the mole concept.
	1.1.A.1: One cannot count particles directly while performing laboratory work. Thus, there must be a connection between the masses of substances reacting and the actual number of particles undergoing chemical changes.
	1.1.A.2: Avogadro's number (NA = 6.022×10^{23} mol^{-1}) provides the connection between the number of moles in a pure sample of a substance and the number of constituent particles (or formula units) of that substance.
	1.1.A.3: Expressing the mass of an individual atom or molecule in atomic mass units (amu) is useful because the average mass in amu of one particle (atom or molecule) or formula unit of a substance will always be numerically equal to the molar mass of that substance in grams. Thus, there is a quantitative connection between the mass of the substance and the number of particles that the substance contains. **EQN: $n = m/M$**
1.3	**Elemental Composition of Pure Substances**
1.3A	Explain the quantitative relationship between the elemental composition by mass and the empirical formula of a pure substance.
	1.3.A.1: Some pure substances are composed of individual molecules, while others consist of atoms or ions held together in fixed proportions as described by a formula unit.
	1.3.A.2: According to the law of definite proportions, the ratio of the masses of the constituent elements in any pure sample of that compound is always the same.
	1.3.A.3: The chemical formula that lists the lowest whole number ratio of atoms of the elements in a compound is the empirical formula.
1.4	**Composition of Mixtures**
1.4A	Explain the quantitative relationship between the elemental composition by mass and the composition of substances in a mixture.
	1.4.A.1: Pure substances contain atoms, molecules, or formula units of a single type. Mixtures contain atoms, molecules, or formula units of two or more types, whose relative proportions can vary.
	1.4.A.2: Elemental analysis can be used to determine the relative numbers of atoms in a substance and to determine its purity.
1.8	**Valence Electrons in Ionic Compounds**
1.8A	Explain the relationship between trends in the reactivity of elements and periodicity.
	1.8.A.1: The likelihood that two elements will form a chemical bond is determined by the interactions between the valence electrons and nuclei of elements.
	1.8.A.2: Elements in the same column of the periodic table tend to form analogous compounds.
	1.8.A.3: Typical charges of atoms in ionic compounds are governed by the number of valence electrons and predicted by their location on the periodic table.
3.7	**Solutions and Mixtures**
3.7A	Calculate the number of solute particles, volume, or molarity of solutions.
	3.7.A.1: Solutions, also sometimes called homogeneous mixtures, can be solids, liquids, or gases. In a solution, the macroscopic properties do not vary throughout the sample. In a heterogeneous mixture, the macroscopic properties depend on location in the mixture.
	3.7.A.2: Solution composition can be expressed in a variety of ways; molarity is the most common method used in the laboratory. **EQN: $M = n_{solute} / L_{solution}$**
3.9	**Separation of Solutions and Mixtures**
3.9A	Explain the results of a separation experiment based on intermolecular interactions.
	3.9.A.1: The components of a liquid solution cannot be separated by filtration. They can, however, be separated using processes that take advantage of differences in the intermolecular interactions of the components.
	3.9A.2: Chromatography (paper, thin-layer, and column) separates chemical species by taking advantage of the differential strength of intermolecular interactions between and among the components of the solution (the mobile phase) and with the surface components of the stationary phase. The resulting chromatogram can be used to infer the relative polarities of components in a mixture.
	3.9A.3: Distillation separates chemical species by taking advantage of the differential strength of intermolecular interactions between and among the components and the effects these interactions have on the vapor pressures of the components in the mixture.

Unit 1: Pre-Review Fundamental Vocab

Types of Particles- Match the definition with the term!

1. **Atom** ________
2. **Molecule** ________
3. **Element** ________
4. **Compound** ________
5. **Ion** ________
6. **Anions** ________
7. **Cations** ________
8. **Diatomic elements** (**Br I N Cl H O F**) ________
9. **Formula Unit** ________

a) particle of an ionic compound; cations and anions bonded together in a neutral compound

b) an atom or molecule that has lost or gained electrons

c) particle with two or more atoms covalently bonded together

d) neutral collection of identical particles containing two or more atoms bonded together

e) a neutral, monatomic particle

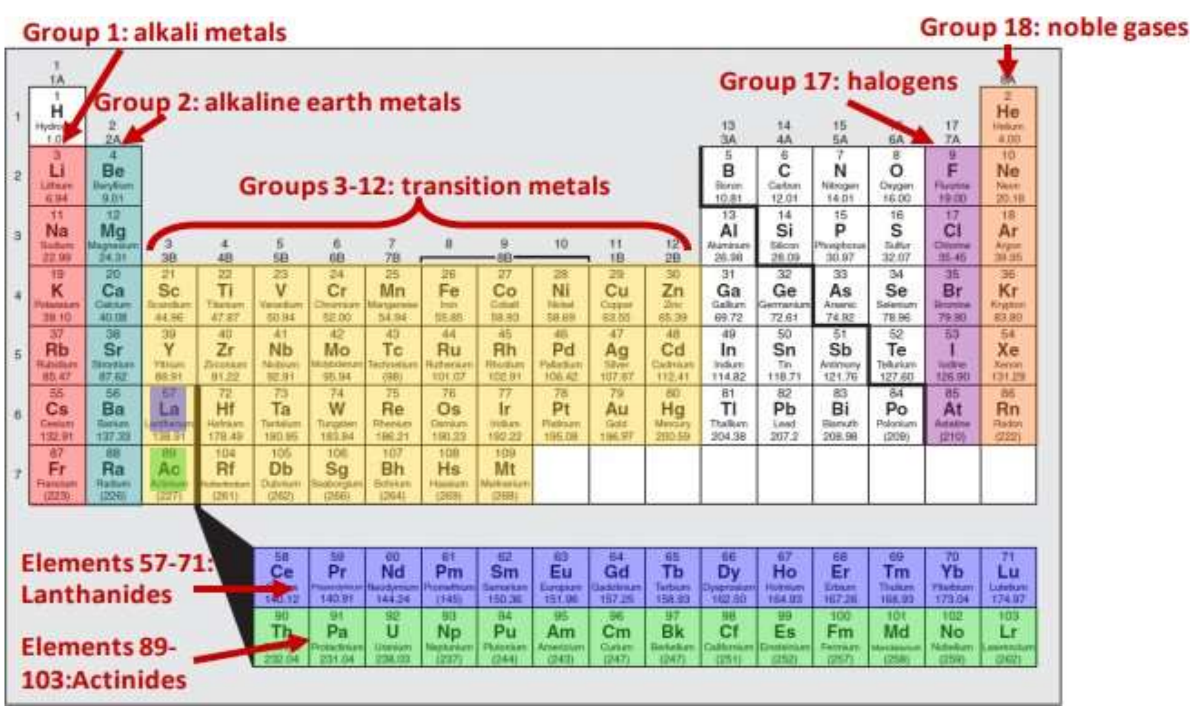
f) positively charged ions (lost electrons)

g) negatively charged ions (gained electrons) ;

h) these elements naturally exist in covalently bonded pairs, they are elemental molecules

i) a substance composed of only one type of atom

You need to know the______________of the periodic table by name, know how many ________________ electrons each family has, and the_____________formed for their common ions! Hint: look at the picture



Year 1 Chem Review Part 1 – Nomenclature

Ionic Nomenclature Review

Ionic Bonding: Metal + Nonmetal(s)

1. Formula to name: Name cation element first (metal), then add the anion element (non-metal) with an –ide ending (but don't change polyatomic ion names)
2. Name to formula: Balance the charges so the total compound adds up to a zero charge.

1) $NaClO_2$ ________________________ 4) barium fluoride ________________________

2) $LiBrO_3$ ________________________ 5) calcium peroxide ________________________

3) $(NH_4)_3N$ ________________________ 6) gallium hydroxide ________________________

Transition metals AND the wanna-be transition metals: **tin (Sn), lead (Pb), antimony (Sb) and bismuth (Bi)**

- Can have more than one charge (**variable charges**) so we use Roman numerals.
- **Exceptions**: Zn^{2+}, Ag^{+}, Cd^{2+} (these do **NOT** use Roman numerals – they only have ONE charge)

Example 1: tin (IV) chloride

Example 2: copper (II) sulfide

Example 3: Fe_2O_3

Example 4: PbS_2

Roman Numerals

1 =

2 =

3 =

4 =

5 =

1) Cu_2O ________________________ 4) scandium (III) chloride ________________

2) MnF_7 ________________________ 5) nickel (II) oxide ________________

Hydrated Ionic Compounds: (hydrates) have a specific number of water molecules in their chemical formula, which is indicated using a Greek prefix.

Greek prefixes									
1	2	3	4	5	6	7	8	9	10
mono	di	tri	tetra	penta	hexa	hepta	octa	nona	deca

1) $Na_2SO_4 \cdot 10H_2O$

3) cobalt (I) chlorate trihydrate

2) $MgCO_3 \cdot 5H_2O$

4) zinc chloride tetrahydrate

Covalent Nomenclature Review

Covalent Bonding: Nonmetal + Nonmetal

1. First element keeps its name, second element gets an –ide ending
2. Use Greek prefixes to identify the number of atoms
3. Exception 1: If there is only 1 first element, no prefix on that element!
4. Exception 2: If there are two of the same element (i.e. diatomics), the first word is element name followed by gas

Note: ***Drop –a and –o endings before "oxide"**

1) As_4O_5 ____________________________________ 5) dihydrogen monoxide ____________________________

2) N_2 ____________________________________ 6) triselenium decoxide ______________________________

3) SO_3 ____________________________________ 7) nitrogen triiodide ________________________________

4) N_2O ____________________________________ 8) sulfur heptafluoride _____________________________

Acid Nomenclature Review

Binary acid contains two different elements: hydrogen and one of the more electronegative elements. Aqueous solutions of these compounds are known by the acid names.

RULES:

1. Name begins with prefix hydro-
2. Root of name of second element follows this prefix
3. Name ends with suffix –ic. And the word acid

EXAMPLE:

1. HBr – hydrobromic acid

Oxyacid is an acid compound of hydrogen, oxygen, and a third element. Usually the oxyacid is one or more hydrogen followed by a polyatomic anion.

RULES:

1. NO PREFIX
2. Look at the polyatomic anion, if the name ended in –ate then the suffix is changed to 'ic' and the word acid is added.
3. If the polyatomic anion was an 'ite', then the suffix is changed to 'ous' and the word acid is added.
4. NO HYDRO prefix!
5. Hint: *Ate-ic ite-ous*

EXAMPLES:

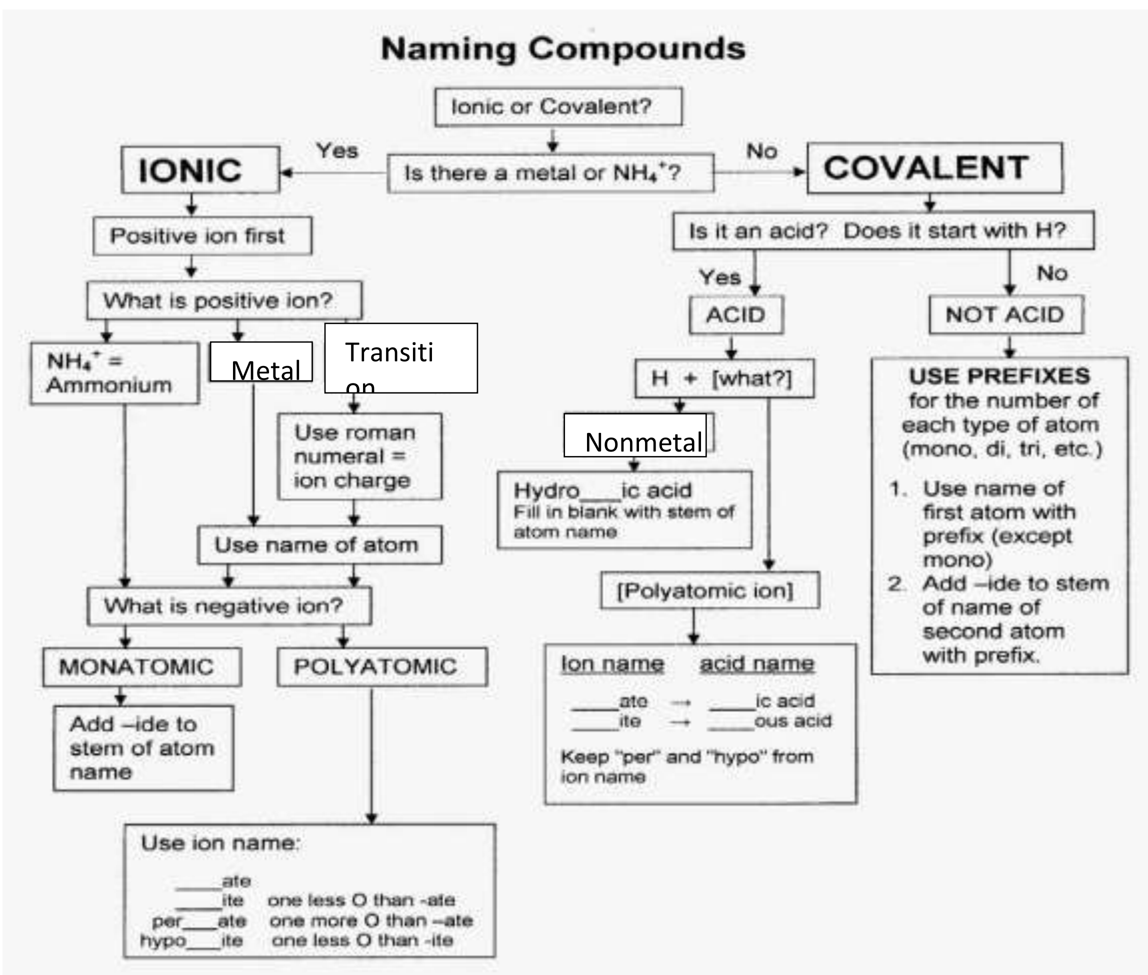
1. H_2SO_4 – sulfuric acid
2. H_2SO_3 – sulfurous acid

Table: Conventions for Naming Oxyacids

Relationship	General name	Example name	Example formula
one more oxygen atom than (root)ic	per(root)ic acid	perchloric acid	$HClO_4$
	(root)ic acid	chloric acid	$HClO_3$
one less oxygen atom than (root)ic	(root)ous acid	chlorous acid	$HClO_2$
two less oxygen atoms than (root)ic	hypo(root)ous acid	hypochlorous acid	HClO

Time to Practice!

1. H_2SO_4 ______________________________
2. HNO_2 ______________________________
3. HCl ______________________________
4. $HBrO_2$ ______________________________
5. nitric acid ______________________________
6. carbonic acid ______________________________
7. hydroiodic acid ______________________________
8. sulfurous acid ______________________________

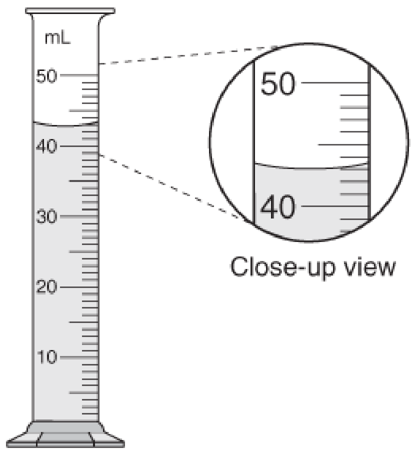


Year 1 Chem Review Part 2: Sig Fig Review

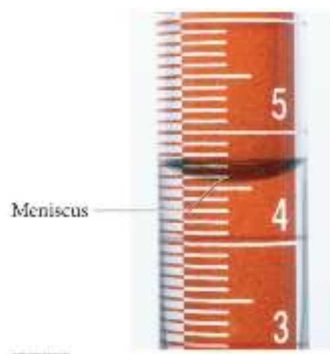
Significant Figures (otherwise and forever known as_____________) in a measurement consist of all the digits known with certainty plus one final digit, which is estimated.

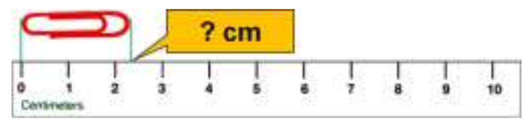
Rule of Thumb: Always measure to the place value measured by the markings on your instrument (e.g. ruler, graduated cylinder) **PLUS ONE MORE.**

Let's Practice! Using correct significant figures, what is the measurement that is represented in each picture?



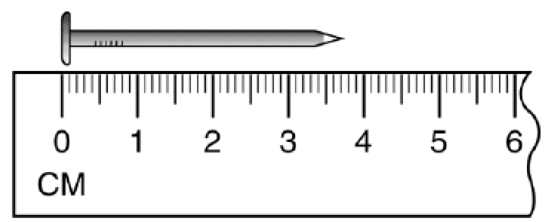
Graduated cylinder





3. Markings?________________

Measurement? ______________



1. Markings?________________

Measurement? ______________

2. Markings?________________

Measurement? ______________

4. Markings? ________________

Measurement? ______________

Rules for Significant Figures

1. Non-zero digits and zeros between non-zero digits are always significant.

   Examples: 102 → ☐sig fig(s), 1.73005 → ☐sig fig(s)

2. Leading zeros are not significant.

   Examples: 0.37 → ☐sig fig(s), 0.0001 → ☐sig fig(s)

3. Zeros to the right of all non-zero digits are only significant if a decimal point is shown.

   Examples: 100 → ☐sig fig(s), 100. → ☐sig fig(s), 0.0100 → ☐sig fig(s)

4. For values written in scientific notation, the digits in the coefficients are significant.

   Examples: 5×10^4 → ☐sig fig(s), 1.30×10^{-13} → ☐sig fig(s)

5. Counting numbers (one kitten, two kittens) and conversion factors (6.022×10^{23} atoms/1 mol) are considered exact value and have an infinite number of sig figs!

   Examples: 3 cats → ☐sig fig(s), 1 student → ☐sig fig(s)

Now You Try!

Number	How many Sig Figs?	Place of significance	Number	How many Sig Figs?	Place of significance
3.0800 mL	5	0.0001	55 puppies		
0.00418 g			1,800. m		
0.0000040 L			1,800 m		
3 people			2.998×10^8 m/sec		

Rules for Sig Fig Calculations (Hint: *Alpha order!* **A***wesome* **P***eople use a* **M***emorization* **T***echnique)*)

1. **A**dding/subtracting: round to least precise **P**lace value of sig fig(A → P) **A***wesome* **P***eople*
2. **M**ultiplying/dividing: round to least precise **T**otal number of sig fig(M → T) **M***emorization* **T***echnique* ☺
3. Note: **do not round any of the numbers you are given until the very end** after you have plugged them into your equations in their full, precise glory! **ALWAYS ROUND LAST!!!**

Let's Try! Addition/Subtraction (Place)

Example #1

2.348
0.07 ←least precise place
+ 2.9975
5.4155 ←calculator answer
5.42 ←rounded answer

Example #2

5.9
− 0.261

Example #3

1,010
2.9
− 0.76

Multiplication/Division (Number)

Example #4: 1.052 x 12.504 x 0.53 = 6.97173024 → **7.0**

(4 sf) ***(5 sf)*** ***(2 sf)*** ***calculator answer*** **rounded answer**

Example #5: 2.0035 ÷ 3.20 =

Example #6: 6.78 x 5.903 x (5.489 – 4.99) =

More Calculations Practice

	Calculator Answer	Rounded Answer (with Correct # of Sig Figs)
1. 170 + 3.5 – 28		
2. 47.0 ÷ 2.2		
3. 691,300 ÷ (5.022 – 4.31)		
4. (0.054 + 1.33) × 5.4		

Year 1 Chem Review Part 3: Atomic Structure and Types of Matter

Coulomb's Law: fundamental relationship between electrostatic________________and_________________.

- It applies to charged particles, magnets, gravitation, etc.
- In chemistry, we are most interested in the______________of attraction or repulsion between __________________ particles

$$E \propto \frac{q_1 q_2}{r}$$

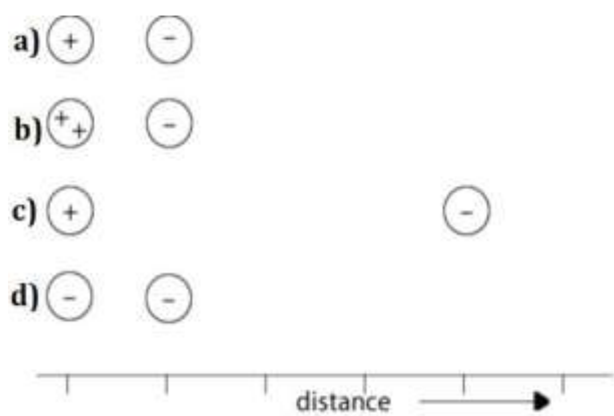
E= energy of attraction or repulsion between particles
q_1 = charge of first particle
q_2 = charge of second particle
r = distance between charged particles

In short:

- Energy of attraction/repulsion_______________as the magnitudes (sizes) of the charges _______________
- Energy of attraction/repulsion_______________as the distance between the charges _______________

Thought question: Will an electron be more attracted to the nucleus of a hydrogen atom or a helium atom, and why?

Examples:



1. Consider the particles in the diagram to the right.
   a. Compare the particles shown in (a) and (b). Which pair is more attracted to each other and why?
   b. Compare the particles shown in (a) and (c). Which pair is more attracted to each other and why?
   c. Compare the particles shown in (a) and (d). Which pair is more attracted to each other and why?

2. An electron would be most attracted to a nucleus containing which of the following?
   a. 7 protons, 5 neutrons
   b. 8 protons, 5 neutrons
   c. 5 protons, 5 neutrons
   d. 7 protons, 8 neutrons

Atomic Theory

- ______________________ is anything that has mass and takes up space.
- All matter is made up of_________________.

Structure of the Atom

- The atom can be divided into two regions: the____________________and the __________________________________.
  1. The___________________is a very small region near the center of an atom that is positively charged.
  2. The_________________________is a very large region that surrounds the nucleus and is negatively charged. It consists mostly of empty space.
- The atom is composed of___________________subatomic particles

The Three Subatomic Particles

Particle	Symbol	Location	Charge	Mass (amu)

- The unit of mass for atomic particles is the____(amu)
  - 1 amu = one-twelfth the mass of a carbon atom containing six protons and six neutrons.

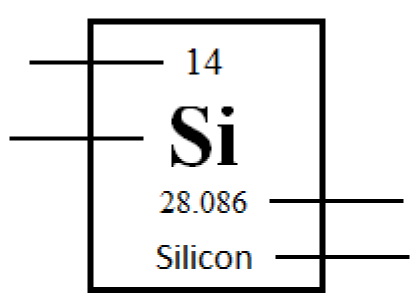
Understanding the Periodic Table

a. **Atomic Number**: the **number** that tells you the_______________of the element; number of_____________

b. **Average Atomic Mass**: average mass of all of the element's_________________
   - To find the mass of a SPECIFIC (isotope) atom, you must add up the protons + neutrons

c. Symbol: shortened element name; starts with a __________________ letter

d. Name: the ___________________________ of an atom (not a proper noun- not capatilized in sentences!



Isotopes

- What are **isotopes**? Atoms of the____________element, but different _____________________
- This means the number of____________________ is the same, and the number of _____________________is different.

→Mass of an isotope = # protons + # neutrons ←

Two ways to write isotopes:

Type	**hyphen-notation**		**isotope notation/ nuclide symbol**	
Definition	name-mass		$^{\text{mass \#}}_{\text{atomic \#}}\text{Symbol}$	
Example	carbon-12	**vs**	$^{12}_{6}C$	
More examples				

Examples: Consider the following sets of isotopes and then explain the similarities and differences between *each* set.

Set I

$^{12}_{6}C$ $^{13}_{6}C$

Similarities:

Differences:

Set II

$^{24}_{12}Mg$ $^{24}_{11}Na$

Similarities:

Differences:

Practice: Complete the following table using your knowledge of atomic structure.

Element	Hyphen notation	Atomic Number	Number of Protons	Number of Neutrons	Number of Electrons	Mass Number
$^{1}_{1}H$						
	chlorine-35					
		27		33		

Classifying Matter

- **Matter**: anything that occupies space and has mass.
- We classify matter according to its **composition** (the basic components that make it up).

Matter

Can it be physically separated?

NO! → **Pure Substances** YES! → **Mixtures**

Pure Substances

Can it be chemically separated into simpler substances?

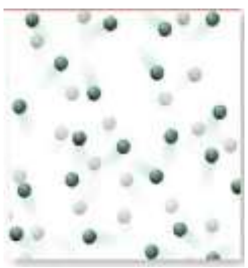
NO! → **Element** YES! → **Compound**

Mixtures

Uniform throughout?

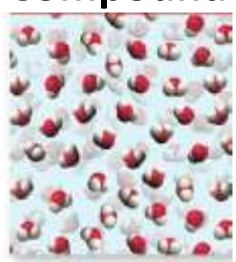
NO! → **Heterogenous Mixture** YES! → **Homogenous Mixture (Solution)**

Element



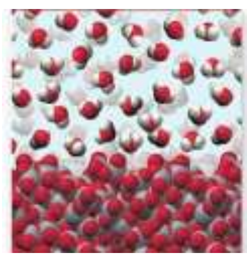
Examples
1.
2.
3.

Compound



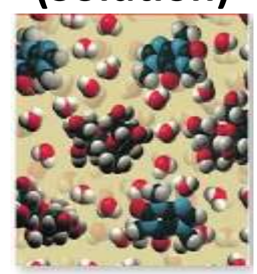
Examples
1.
2.
3.

Heterogenous Mixture



Examples
1.
2.
3.

Homogenous Mixture (Solution)



Examples
1.
2.
3.

Pure Substances - Must be separated chemically (bonds broken)

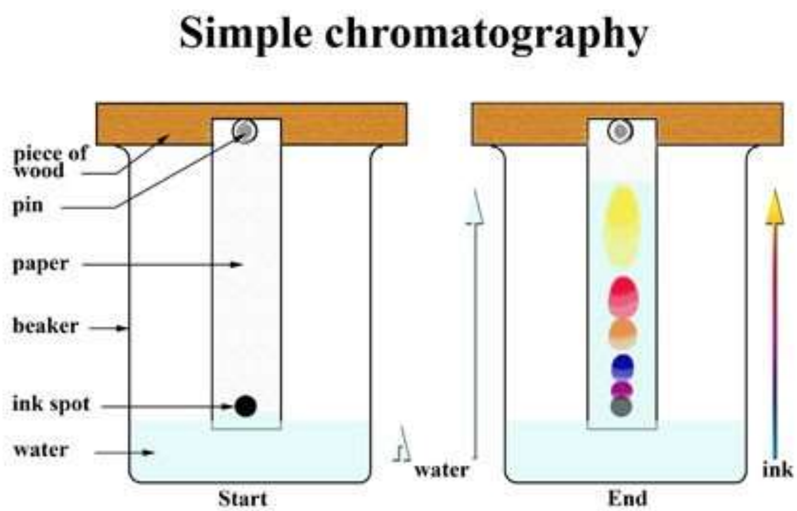
- Same/fixed ______________________________
- **Elements**
  - Cannot be broken down and still maintain identity
  - One type of atom (basic building blocks)
  - Found on the Periodic Table!
- **Compounds**
  - Chemical combination of____________or more elements in fixed, definite proportions
  - Cannot be separated by physical means
  - Properties of compound are different than individual elements

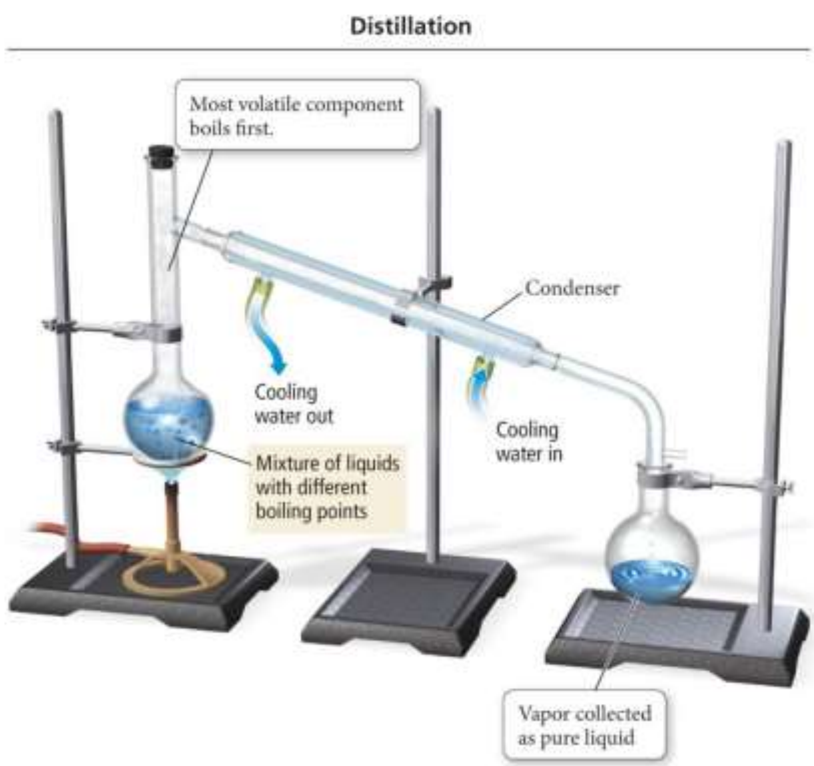
Mixtures - Can be separated by physical means

- Formed when two or more substances (s, l, g, aq) are physically combined
- Has more then one ________________ type____________________
- All substances in mixture retain their own properties
- **Heterogenous**
  - Parts of the mixture are NOT evenly distributed (poorly mixed)
  - Does_____________look the same throughout
- **Homogenous**
  - Parts of the mixtures are____________________distributed (evenly mixed)
  - Also called a ____________________

Methods for Separating a Mixture: Both heterogeneous and homogeneous mixtures can be separated by ________________ means into the component parts that make up the mixture.

1. A solid and liquid mixture can be separated by pouring the mixture through a_____________paper designed to allow only the liquid to pass.
2. A homogeneous mixture of liquids can be separated using____________________, a process in which the mixture is heated and the more volatile (more easily vaporized) liquid is boiled off first. A condenser is then used to recollect the vaporized component.
3. Paper________________________takes advantage of the fact that different components of a homogeneous mixture have different attractions to a solvent and paper.





Year 1 Chem Review: All Together Practice!

Name/Write formulas for the following compounds.

Formula	Type of bond (ionic/covalent)	Name
Fe_2S_3		
$AlPO_4 \cdot 7H_2O$		
Si_2Br_6		
NaBr		
C_3H_8		
Na_2SO_4		
$Co_3(PO_4)_2$		
CF_4		
$Ca(NO_2)_2$		
N_2O_3		
VO_2		
$LiBr \cdot 4H_2O$		
SrO		
Ag_3P		
H_2SO_4		
		cadmium (II) nitride
		tetraphosphorus triselenide
		copper (IV) phosphate pentahydrate
		zinc oxide
		hydrophosphoric acid
		aluminum chloride
		potassium carbonate
		perbromic acid
		dinitrogen trioxide
		nickel (III) sulfide
		tin (IV) chlorate
		iodine pentafluoride
		ammonium oxide
		boron trifluoride

1. A halide is a binary compound, of which one part is a halogen atom and the other part is an element or radical that is less electronegative than the halogen, to make a fluoride, chloride, bromide, iodide or astatide compound. What is the general formula for alkali metal halides?

   a. MX b. M_2X c. MX_2 d. MX_3

2. Which electrons account for many of the chemical and physical properties of elements?

   a. Innermost b. Intermediate c. Outermost d. Transition

3. Which of the following elements is not correctly paired with its group (family) name?

   a. Bismuth (Bi), halogens c. Lithium (Li), alkali metals

   b. Strontium (Sr), alkaline earth metals d. Radon (Rn), noble gases

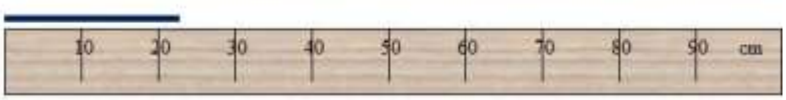
4. The inertness of noble gases is due to:

   a. The number and arrangement of their electrons
   b. The unique structure of their nuclei
   c. The special number of protons and neutrons
   d. The bonds they form with other elements

Sig Fig Practice

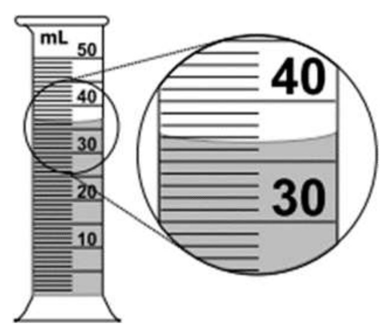
Directions: Measure the following to the correct number of significant figures.

1. ____________



2. ____________

3. ____________



4. ____________

Directions: Determine the number of significant figures.

1. 0.02_________ **2.** 142 ________ **3.** 0.02020 ________ **4.** 0.073 ________

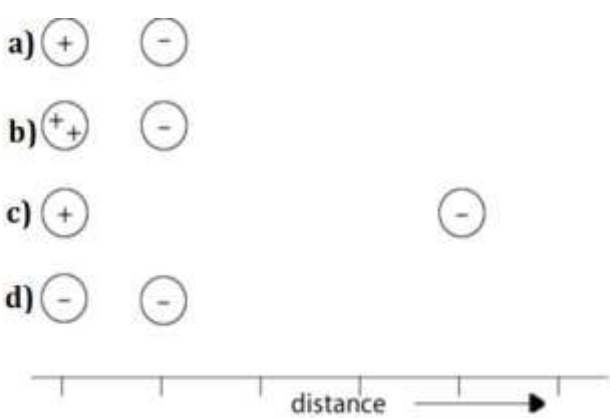
5. 501_________ **6.** 1.071 ________ **7.** 501.0 ________ **8.** 10810 ________

9. 5000_________ **10.** 5.00 ________ **11.** 5000. ________ **12.** 1.20×10^3________

Directions: Round your answer to the correct number of sig figs.

	Calculator Answer	**Rounded Answer** (with Correct # of Sig Figs)
1. 140 cm × 35 cm		
2. 0.003 L + 0.0048 L + 0.100 L		
3. 67.35 cm^2 ÷ (1.401 cm – 0.399 cm)		
4. (6.23 cm + 3.111 cm – 0.05 cm) × 14.99 cm		
5. 3.14159 m × (4.17 m + 2.150 m)		

Coulomb's Law



1. Which set of particles shown the left will experience:
   a. the greatest attraction to each other? _______
   b. the greatest repulsion from each other? ______
2. The particles in (a) will experience_______________attraction to each other than the particles in (c) because:
   a. greater, the distance between them is less.
   b. smaller, the distance between them is less.
   c. the same, distance is irrelevant to force of attraction.
3. The particles shown above in (a) will experience_______________attraction to each other than the particles in (b) because:
   a. greater, the nucleus has one proton instead of two.
   b. smaller, the nucleus has one proton instead of two.
   c. the same, charge is irrelevant to force of attraction.

4. An electron in the lowest energy level would be most attracted to the nucleus of which element?
   a. lithium
   b. sodium
   c. potassium
   d. rubidium

5. A proton would be least repulsed by the nucleus of which element?
   a. helium
   b. neon
   c. argon
   d. krypton

Isotopes! Complete the chart below.

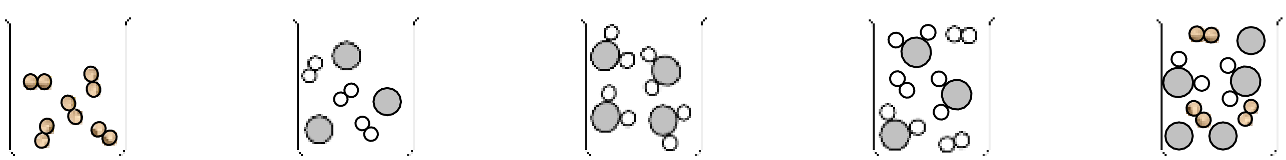
Hyphen notation	Isotope Notation	Mass #	Atomic #	Protons	Neutrons	Electrons
	${}^{10}_{5}B$					
phosphorus–31						
		66		30		
			27		32	

Types of Matter

Directions: Identify the following as: element (E), compound (C), heterogeneous mix (He) or homogeneous mix (Ho).

a. Table salt __________
b. Nitric acid (HNO_3) __________
c. Sugar (Glucose) __________
d. Carbon dioxide (CO_2) __________
e. Milk __________
f. Air __________
g. Nitrogen gas (N_2) __________
h. Zinc (Zn) __________
i. Pulpy orange juice __________

Directions: Use the particle representations below to answer #1-5. **Answer choices may be used more than once!**



A B C D E

1. Compound ______
2. Nitrogen, N_2 ______
3. Mixture of two elements ______
4. Element_______
5. Mixture of water, H_2O, and hydrogen H_2 ________

Directions: Choose the best possible answer for each question.

1. A forensic technician is examining a fine white powder found at a crime scene. It appears to be of uniform texture and consistency. He finds that upon heating, the substance decomposes, releasing a colorless gas and leaving a black residue. Based on these observations, the substance is:

A a homogeneous mixture
B a heterogeneous mixture
C either a homogeneous mixture or a compound
D a compound

2. An example of a heterogeneous mixture is

A bronze. **B** concrete. **C** brass. **D** water.

3. How is a mixture different from a compound?

A Particles of a mixture are combined chemically.
B Components of a mixture can only be separated chemically.
C Components of a mixture can be separated by physical means.
D Composition of a mixture may be constant.

4. A transparent liquid has uniform color. Under a microscope, no difference in uniformity is apparent. The liquid all boils away at the same temperature. Based on this information, it can be concluded that the liquid is:

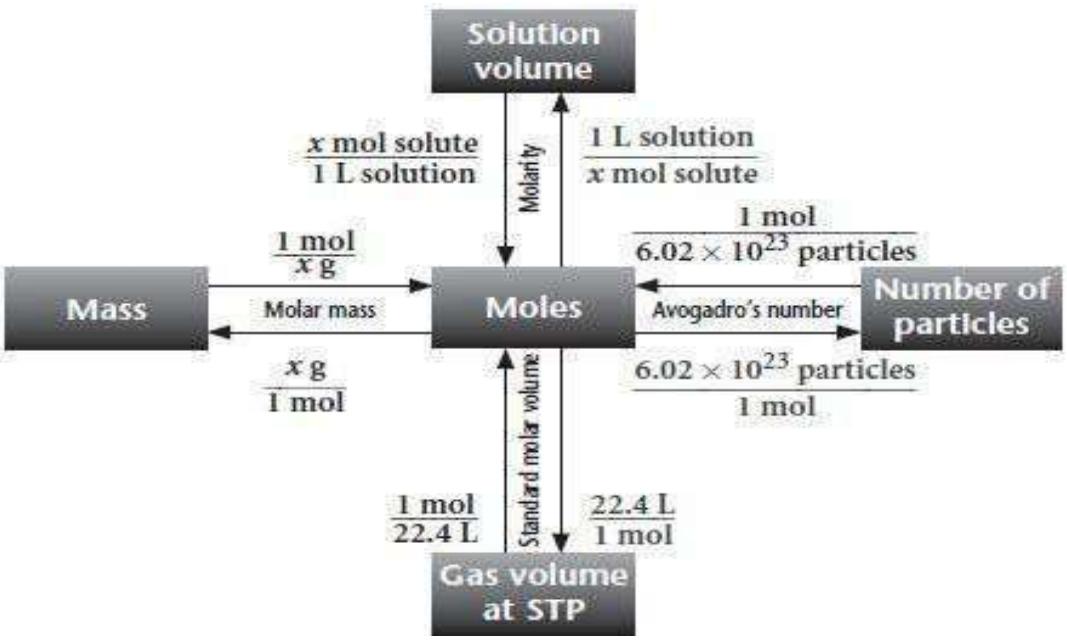
A an element
B a homogeneous mixture
C either a pure substance or a homogeneous mixture
D a pure substance

Here and Back Again: A Mole-ish Review

Part I: The Mole – Massively Important

- The **mole** is a unit of counting used in chemistry.________________**number** (6.022 x 10^{23}) represents the number of___________________(atoms, ions, formula units, or molecules) in one mole of any substance.
- Any atom, element, or compound can have its mass expressed in atomic mass units (_______). The average atomic mass (in amu) for atoms of any element can be found on the **periodic table**.
- Atomic mass units can be_______________to find the corresponding mass (again in amu) of any given multi-atom species.
- The numerical value of the **amu** of the atoms of a given element is equal to the **mass** in_________of one **mole** of that element.
- This allows the conversion of the number of **particles** (atoms, ions, formula units, or molecules), the number of **moles**, and the **mass** of any given substance.

* Remember: one_______________is one covalent compound unit, while one formula unit is one ionic compound unit.



1. Freon, which has the formula CCl_2F_2, is used as a refrigerant in air conditioners and as a propellant in aerosol cans. Given a 5.56 mg sample of Freon, calculate the number of molecules of freon in that sample.

2. If two atoms of carbon combine with four atoms of hydrogen in the compound ethene (C_2H_4), how many grams of hydrogen would be needed to combine completely with 6.0 grams of carbon? Let's try this one without a calculator!

Part II: The Mole – A Solution to Every Problem

- A____________________is a homogeneous mixture: the properties are the same no matter what part of the sample one examines.
- Solution concentration is usually expressed in terms of______________(M), i.e., the number of **moles** of solute per **liter** of solution.

$$M = \frac{mol}{L}$$

- Molarity has the units of______,_________________, or:

3. You prepare a solution by dissolving 48.05 g of $MgBr_2$ in enough water to make 800. mL of solution.

a. Calculate the molarity of Mg^{2+} and Br^-.

b. How many bromide ions are found in this solution?

c. What fraction of the total number of ions are bromide ions?

4. How many OH^- ions are contained in 2.5 L of a 0.52 M $Al(OH)_3$ solution?

Part III: The Mole – It's Such a Gas!

- At________, **one mole** of any ideal gas occupies a volume of______________.
  - STP: Standard temperature is 0°C (273K) and standard pressure is 1 atm = 760 mm Hg = 760 torr.
- Gases must have their quantitative properties calculated using temperatures in_________________.
- The ideal gas equation,_____________________, can be used to relate the properties [pressure (P), volume (V), number of moles (n), and *Kelvin* temperature (T)] of ideal gases to one another.
  - R = universal gas constant = 8.314 J mol^{-1} K^{-1} = 0.08206 L atm mol^{-1} K^{-1} = 62.36 L torr mol^{-1} K^{-1}
  - The numerical value of R is *dependent upon the units used*.
- A rhyme to help you remember! **When you're_______at STP, use PV = nRT.**

5. Given 7.81 x 10^{22} molecules of chlorine gas (Cl_2):

a. What is the volume of your sample at STP?

b. What is the volume of your gas sample in Texas in late August? Let's say 1.00 atm and 38°C (about 100°F, sigh).

6. Three identical containers are filled with a single gas each: the first container with O_2, the second with CO_2, and the last with N_2. All of the containers are at the same temperature and pressure.

a. Which container has the greatest number of molecules?

b. Which container has the smallest mass?

Mole Review: Multiple Choice Practice Problems

Fun and exciting note about AP Chem: did anyone mention that you're NOT allowed to use a calculator on the multiple choice portion of the AP Chem test? It's true! <cue applause>

To help you be prepared for the AP test in May, all of your quizzes and tests in AP Chemistry will have two sections: a multiple choice section (no calculator) and a free response section (with a calculator). But don't worry: throughout the year, we're going to teach you lots of tips and tricks to make calculator-free math pHun and delicious.

Guided Practice

1. A single sodium atom has an average mass of 22.99 amu (taken from the periodic table). How does this number relate to a mole of sodium atoms?

a. One mole of sodium atoms will have a mass of 22.99 amu.

b. One mole of sodium atoms will have a mass of 22.99 grams.

c. One mole of sodium atoms will have a mass of 22.99 x 6.022 x 10^{23} grams.

d. You cannot relate the mass of a single sodium atom to the mass of a mole of atoms.

2. How many protons are in 3.50 moles of lithium atoms?

a. 3.50 b. 10.5 c. 3.50 x (6.022 x 10^{23}) d. 10.5 x (6.022 x 10^{23})

3. Which of the following samples contains the largest number of atoms?

a. 3.0 mol of water b. 3.0 g of water c. 3.0 x 10^{22} molecules of nitrogen gas

Independent Practice

4. What volume will 37.8 g of oxygen gas occupy at STP?

a. 847 L b. 52.9 L c. 26.5 L d. 1.69 L

5. At 0°C and 1 atm, a 0.25 L container would hold how many grams of carbon monoxide? (The molar mass of carbon monoxide is 28.01 g/mol).

a. 7.0 g b. 3.2 g c. 0.31 g d. 0.011 g

6. If you require 13.9 grams of lithium atoms, how many grams of lithium sulfide would be required?

a. 27.8 g b. 39.1 g c. 46.0 g d. 92.0 g

7. Calculate the concentration of a solution prepared by dissolving 11.85 g of solid $KMnO_4$ in enough water to make 750. mL of solution. (The molar mass of $KMnO_4$ is 158.04 g/mol).

a. 0.100 M b. 0.0562 M c. 1.00 M d. 0.562 M

8. How many moles of a gas at 247°C would occupy a volume of 6.4 L at a pressure of 260 mmHg?

a. 0.051 mol b. 0.11 mol c. 0.28 mol d. 3.9 mol

9. How many total ions are contained in 250 mL of a 0.100 M NaCl solution?

a. 1.5×10^{22} ions b. 3.0×10^{22} ions 1.5×10^{25} ions b. 3.0×10^{25} ions

10. What is the molar concentration of $Al(OH)_3$ if a 500. mL solution of $Al(OH)_3$ contains 1.8×10^{24} ions of OH^-?

a. 0.020 M b. 0.060 M c. 2.0 M d. 6.0 M

11. What is the total number of atoms in 123 grams of sulfur trioxide?

a. 9.25×10^{23} atoms b. 3.70×10^{24} atoms c. 5.93×10^{27} atoms d. 2.37×10^{28} atoms

Below in an example of a Composition of Mixtures problem. In these problems, a sample contains more than one substance. By isolating one of the substances in a mixture, the percent composition by mass can be determined. In these problems, it is important to understand that the total sample in a mixture, and cannot be analyzed as a whole in terms of moles. Unlike a pure substance, which has an empirical formula, a mixture has an indefinite proportion of elements. Be careful to recognize a mixture and analyze it carefully!

Mass of KI tablet	0.425 g
Mass of thoroughly dried filter paper	1.462 g
Mass of filter paper + precipitate after first drying	1.775 g
Mass of filter paper + precipitate after second drying	1.699 g
Mass of filter paper + precipitate after third drying	1.698 g

1. A student is given the task of determining the I^- content of tablets that contain KI and an inert, water-soluble sugar as a filler **(how does this statement show that this is a mixture problem?)**. A tablet is dissolved in 50.0 mL of distilled water, and an excess of 0.20 *M* $Pb(NO_3)_2(aq)$ is added to the solution. A yellow precipitate forms, which is then filtered, washed, and dried. The data from the experiment are shown in the table above.

   (a) For the chemical reaction that occurs when the precipitate forms,

      (i) write a balanced, equation for the reaction indicating the phase (*s, l, g, or aq*).

      (ii) What substance or substances are being collected in the filter paper?

   (b) Explain the purpose of drying and weighing the filter paper with the precipitate three times.

   (c) In the filtrate solution, is $[K^+]$ greater than, less than, or equal to $[NO_3^-]$? Justify your answer.

   (d) Calculate the number of moles of precipitate that is produced in the experiment.

   (e) Calculate the mass percent of I^- in the tablet.

   (f) In another trial, the student dissolves a tablet in 55.0 mL of water instead of 50.0 mL of water. Predict whether the experimentally determined mass percent of I^- will be greater than, less than, or equal to the amount calculated in part (e). Justify your answer.

Percent Composition by Mass

Percent Composition: the percent by_____________of each element in a compound.

- According to the law of____, a given chemical compound always contains the exact same elements in the exact same ratio by______________.

$$\% \text{ composition of an element } = \frac{\text{total mass of the element in the compoung}}{\text{total mass of the compound}} \times 100$$

Guided Practice

1. Find the percentage composition of each element in the compound copper (I) sulfide.

2. Find the mass percentage of water in sodium carbonate decahydrate, $Na_2CO_3 \bullet 10H_2O$, which has a molar mass of 286.15 g/mol.

3. When ammonia, NH_3, is formed, 1.0 gram of hydrogen reacts with about 5.0 grams of nitrogen. How much nitrogen would be needed to react with 2.5 grams of hydrogen in the production of ammonia?

4. Lake Superior is the largest lake in North America and contains about 1.2 x 10^{16} kg of water. What mass of hydrogen is contained in Lake Superior?

   a. 6.0 x 10^{15} kg
   b. 1.1 x 10^{16} kg
   c. 1.3 x 10^{15} kg
   d. 1.2 x 10^{16} kg

5. For a 150 g sample of glucose, $C_6H_{12}O_6$, there is 60 g of carbon. How many grams of carbon are there for a 300 g sample of glucose?

   a. 30 g
   b. 60 g
   c. 90 g
   d. 120 g

Empirical and Molecular Formulas

Empirical Formula: the symbols for the__________________combined in a compound, with __________________ showing the smallest whole-number mole ratio of the different atoms in the compound.

Molecular Formula: the______________formula of a compound which shows the total number of each atom in the molecule.

*** It is possible for the empirical formula and the molecular formula to be the______________! ***

Molecular	Simplify by dividing	Empirical
$C_6H_{12}O_6$	6	CH_2O
P_4O_{10}		
	3	C_3HON_4
NaCl		

Steps for the Calculation of Empirical Formula

Step 1: Convert % composition to a mass composition by assuming 100 g of sample.

Step 2: Convert the mass into moles. **DO NOT ROUND!! Keep at least 3 #.**

Step 3: Divide by each number of moles by the smallest number of moles calculated in step #2.

Step 4: If not a whole number, multiply to get rid of the decimal (see chart on right) and use those whole numbers as subscripts in compound.

Fractional Subscript	Multiply by This
0.20	5
0.25	4
0.33	3
0.40	5
0.50	2
0.66	3
0.75	4
0.80	5

The Empirical Song! *(sung with the melody from Twinkle, Twinkle Little Star)*

Percent to mass and mass to mole,

Divide by small then multiply til whole.

That's how you find the empirical

Smallest whole-number ratio.

Let's Practice!

1. Analysis of a sample of a compound known to contain only phosphorus and oxygen indicates that it contains 43.67% phosphorus by mass. What is the empirical formula and name of this compound?

How to Determine the Molecular Formula

Step 1: Find the mass (formula weight) of the empirical formula.

Step 2: Take the molecular mass and divide it by empirical mass (this will always give you a whole number).

Step 3: Multiply the whole # by the empirical formula's subscripts to determine the molecular formula.

Let's Practice!

2. What is the molecular formula for a compound with the empirical formula H_2O and a molecular mass of 54 g/mol?

3. A sample of NutraSweet is 57.14% C, 6.16% H, 9.52% N, and 27.18% O. Calculate the empirical formula of NutraSweet and find the molecular formula. (The molar mass of NutraSweet is 294.30 g/mol)

4. The analysis of a rocket fuel sample showed that it contained 87.4% nitrogen and 12.6% hydrogen by weight. Mass spectral analysis showed the fuel to have a molar mass of 32.06 g/mol. What is the molecular formula of the fuel?

Multiple Choice Practice

1. An element X combines with oxygen to form a compound of formula XO_2. If 24.0 g of element X combine with exactly 16.0 g of O to form this compound, what is the atomic weight of element X?

   a. 48.0 amu b. 24.0 amu c. 16.0 amu d. 12.0 amu

2. A new ore contains 52.3% silver by mass. How many grams of the ore are needed to obtain 10.0 moles of silver?

   a. 2,060 g b. 1,080 g c. 564 g d. 10.0 g

3. A compound is made up of entirely silicon and oxygen atoms. If there are 14.0 g Si and 32.0 g O present, what is the empirical formula of the compound?

   a. SiO_2 b. SiO_4 c. Si_2O d. Si_2O_3

4. A chemist suspects that a given sample of $CaSO_4$ is impure. Upon testing, the chemist finds that the sample contains 31.6% calcium, but pure $CaSO_4$ is 29.4% calcium by mass. Which of the following might account for the measured percent mass of the sample?

   a. The sample is composed entirely of $CaSO_4$.

   b. The sample is a mixture of $CaSO_4$ and $CaCO_3$.

   c. The sample is a mixture of $CaSO_4$ and $MgSO_4$.

   d. The sample is a mixture of $CaSO_4$ and $Ca(BrO_3)_2$.

5. Styrene has the empirical formula CH with a molar mass of 104.13 g/mol. Approximately how many hydrogen atoms are present in a 52 g sample of styrene?

   a. 5.0×10^{22} H atoms
   b. 2.4×10^{24} H atoms
   c. 8.0×10^{23} H atoms
   d. 6.1×10^{23} H atoms

6. What is the empirical formula for a compound that contains 7.48 g N and 1.08 g H?

   a. N_2H
   b. NH_2
   c. NH
   d. NH_5

7. If the compound in #6 has a molecular mass of 32.05 g/mol, what is its molecular formula?

   a. N_2H_2
   b. N_4H_2
   c. N_2H_{10}
   d. N_2H_4

8. An Olympic medal contains 71.5% of gold by mass. How much gold could be extracted from a medal that weighs 115 g?

   a. 0.417 mol
   b. 0.817 mol
   c. 2.43 mol
   d. 4.86 mol

9. Two different samples are analyzed. Sample A contains 2.8 g of N and 1.6 g of O. Sample B contains 14.0 g of N and 8.0 g of O. Which of the following statements is true?

   a. Sample A and B are the same compound because they contain the same types of atoms.
   b. Sample A and B are the same compound because their mass ratios indicate they both contain the same ratio of atoms within the molecule.
   c. Sample A and B are different compounds because they contain different numbers of atoms indicating a different ratio of atoms within the molecule.
   d. Sample A and B are different compounds because their molar masses are different.

Hydrates: Salty salts with a hidden surprise!

A **<u>hydrate</u>** is a_________substance (often ionic) that contains a fixed composition of water molecules (known as "waters of hydration") embedded in its crystal structure.

- ➔ Heating a hydrate "drives off" the water molecules, and the solid that remains behind is called anhydrous, meaning "without water."
- ➔ By measuring the mass of water removed when dehydrating a hydrate, we can determine the ratio of water molecules to anhydrous salt for a given hydrate, which allows us to find the formula of the hydrate!

Notes about Language: Talking about hydrates can be tricksy! Here's a quick guide to the terminology used.

Word/ Phrase	Meaning/ Context
Hydrate	Pure substance, typically crystalline, containing a fixed ratio of water molecules within its structure → this term is only used *before* water is removed by heating!
Water molecules "driven off"	The process of forcing out the embedded water molecules in a hydrate through heating (<u>do NOT call this evaporation</u> – different context!)
Anhydrous salt	Compound remaining after all water molecules have been 'driven off' (removed)

Hydrate Math

The percent of water in a hydrate can be determined in a manner similar to determining the percent composition of a compound.

$$\% H_2O = \frac{\text{Mass of water}}{\text{Mass of hydrate}} \times 100\%$$

Steps to gravimetrically (by mass) **determine the formula of a hydrate:**

1. Determine the_________of the water that has left the compound.
2. Convert the mass of water to___________.
3. Convert the mass of anhydrate that is left over to moles.
4. Find the water-to-anhydrate mole________(just like finding an___________________formula, but be careful: you can't multiply til whole! The mole ratio of the anhydrous salt must always be_____; only the number of waters can be a whole number greater than 1)
5. Use the mole ratio to write the formula.

Example: The following data were obtained when a sample of barium chloride hydrate was analyzed:

Mass of empty test tube	18.42 g
Mass of test tube and hydrate (before heating)	20.75 g
Mass of test tube and anhydrous salt (after heating)	20.41 g

a. Calculate the mass of water lost from the hydrate.

b. Calculate the percentage of water in the original sample.

c. Calculate the moles of water lost from the sample.

d. Calculate the moles of anhydrous salt remaining after the sample was heated.

e. Determine the formula for the hydrate.

Multiple Choice Practice:

1. A sample of a hydrate of $CuSO_4$ with a mass of 250 grams was heated until all the water was removed. The sample was then weighed and found to have a mass of 160 grams. What is the formula for the hydrate?

   a. $CuSO_4 \cdot 10\ H_2O$ b. $CuSO_4 \cdot 7\ H_2O$ c. $CuSO_4 \cdot 5\ H_2O$ d. $CuSO_4 \cdot 2\ H_2O$

2. The anhydrous salt X_2CO_3 has a molar mass of 106 g/mol. A hydrated form of this salt is heated until all of the water is removed and it loses 54% of its mass. The formula of the hydrate is:

   a) $X_2CO_3 . 7\ H_2O$ b) $X_2CO_3 . 5\ H_2O$ c) $X_2CO_3 . 3\ H_2O$ d) $X_2CO_3 . H_2O$

Common Lab Errors when Determine the Formula of a Hydrate

Error	**Effect on Calculated % H_2O**
Excess________________caused the dehydrated sample to decompose. • Often times, a gas will be released during the decomposition	• Gas from the decomposition will be lost *as well as* the expected water loss from heating the hydrate. • The calculated % H_2O will be______________than the actual % H_2O in the hydrate.
The dehydrated sample absorbed moisture from the air after heating (but before the mass is measured).	• Not all of the waters of hydration will be removed. • The calculated % H_2O will be______________than the actual % H_2O in the hydrate.
The hydrate is not heated to "_________________mass" • The hydrate should be heated multiple times and the mass measured each time, to ensure all of the water molecules have been driven off.	• Not all of the water molecules will have been driven off, so the remaining salt is not completely anhydrous. • The calculated % H_2O will be______________than the actual % H_2O in the hydrate.

Practice with Hydrates: Thirst-Quenching!

Error Analysis Practice:

1. A student wants to experimentally determine the number of moles of water in one mole of $BeC_2O_4 \cdot 3\ H_2O$. After heating the sample and measuring the new mass, the student calculated the amount of water driven off. Using that value and her experimental data, she derives the formula of the hydrate as $BeC_2O_4 \cdot 2\ H_2O$. Provide a reasonable explanation for an error that might have caused this outcome and explain how this error affected the student's results.

2. Another student repeats the attempt to experimentally derive the empirical formula of of $BeC_2O_4 \cdot 3\ H_2O$. He decides to heat the sample multiple times over a higher heat to be certain to drive off all water from the hydrate sample. Using his data, this student calculated the formula of the hydrate to be $BeC_2O_4 \cdot 4\ H_2O$. Explain what error might have resulted in this outcome.

AP Free Response

or How I Learned to Stop Worrying and Love AP Chemistry

The AP Chemistry test absolutely love love loves to give you very long questions, where many of the answers rely on answers determined in earlier parts of the question. This has stymied many a determined AP chemistry student in the past and can significantly lower your overall score. You need to know how to best show AP chemistry graders (and me!) all of the chemistry you know as quickly as possible, which means not getting stuck on one tiny part of a problem.

<u>Question</u>: What do you do when you know exactly how to answer parts d. – g., but you have no idea how to get the answer to part c. (which you need for the rest of the question)?

<u>Answer</u>: Make up an answer! For reals. The AP Chemistry scorers will give you FULL POINTS if you correctly show your work and chemistry knowledge on a question part, *even* if your final answer is wrong because you started with an incorrect value from a previous wrong answer. (The wrong calculation will still be counted wrong, as it should be.)

AP Free Response Sample Question (10 points)

1. In the laboratory, a sample of pure nickel was placed in a clean, dry, weighted crucible. The crucible was heated so that the nickel would react with the oxygen in the air. After the reaction appeared complete, the crucible was allowed to cool and the mass was determined. The crucible was reheated and allowed to cool. Its mass was then determined again to be certain that the reaction was complete. The following data was collected during the experiment. Use this data to answer the questions below.

Data Table 1: Calculation of Empirical Formula of Nickel Oxide

Mass of crucible	30.02 g
Mass of nickel and crucible	31.07 g
Mass of nickel oxide and crucible	31.36 g

a. What is the mass of nickel used? [1 point]

b. What is the mass of nickel oxide produced? [1 point]

c. What mass of oxygen reacted? [1 point]

d. Based on your calculations, what is the empirical formula for the nickel oxide? [2 points]

e. Based on your knowledge of reduced layered perovskite synthesis, what is the molecular weight of the compound? [2 points]

f. Determine the molecular formula of the compound. [1 point]

g. While waiting for the sample to cool, the student begins to clean up the lab station using a nearby sink. If, unknown to the student, some water splashed into the crucible, how would this affect the calculated empirical formula? [1 point]

Combustion Analysis

<u>Combustion Analysis</u>: Technique used to obtain the empirical formula of a ____________________

➔ Remember a standard (unbalanced) combustion reaction? (This formula is unbalanced!)

$$C_xH_yO_z + O_2 \rightarrow \_\_\_\_\_\_\_ + \_\_\_\_\_\_\_$$

How to Solve a Combustion Analysis Problem

1. Convert mass of CO_2 and mass of H_2O to____________of each compound.
2. Convert moles of CO_2 to moles of________________, and moles of H_2O to moles of________________.
3. **<u>If</u>** the compound contains something which is__________C or H, find its mass by subtraction, and convert the mass to moles.
4. Now you have mole numbers! Complete the______________formula calculation (divide by small, multiply til whole).

Practice:

1. Upon combustion, a 0.8233 g sample of a compound containing only carbon, hydrogen, and oxygen produces 2.445 g CO_2 and 0.6003 g H_2O. What is the empirical formula of the compound?

Practice Makes Perfect!

2. Combustion analysis determined that a compound containing only carbon and hydrogen produces 1.83 g CO_2 and 0.750 g H_2O. The molar mass of the substance is 42 g/mol. Find the empirical and molecular formulae of the compound.

3. When the unbalanced reaction below occurs at STP, 1.5 L of CO_2 and 1.0 L of H_2O are created. What is the empirical formula of the hydrocarbon?

$$C_xH_y(g) + O_2 \rightarrow CO_2(g) + H_2O(g)$$

a. CH_2 b. C_2H_3 c. C_2H_5 d. C_3H_4

4. Combustion analysis of 0.800 g of an unknown hydrocarbon yields 26 g CO_2 and 7.8 g H_2O. What is the formula of the hydrocarbon?

a. CH_2 b. C_2H_3 c. C_2H_5 d. C_3H